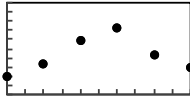
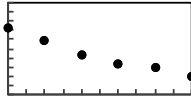
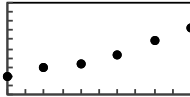
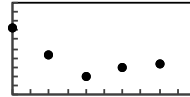


1. Partially because of people from a certain country living longer, the number with Alzheimer's disease and other dementia is projected to grow each year. The table below gives the millions of citizens in the country with Alzheimer's from 2000 and projected to 2050. Complete parts (a) through (e).

Year	2000	2010	2020	2030	2040	2050
Number	3.9	5.7	6.6	8.8	11.8	14.4

a. Create a scatter plot of the data, with x equal to the number of years after 2000 and y equal to the number of millions of citizens with Alzheimer's disease or other dementia. Choose the correct graph below.

☐ A.☐ B.☐ C.☐ D.

All graphs are shown in $[0, 50]$ by $[0, 20]$ with $X_{\text{scl}}=5$ and $Y_{\text{scl}}=2$ viewing rectangles.

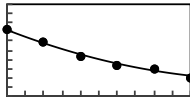
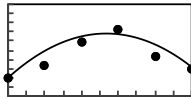
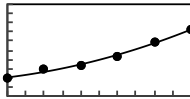
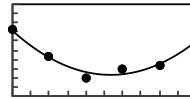
b. Find a quadratic function that models y , the number of millions of people with Alzheimer's disease or other dementia, as a function of x , the number of years after 2000. Report the model with four significant digits.

$$y = (\text{ })x^2 + (\text{ })x + (\text{ })$$

(Type integers or decimals.)

c. Graph the data and the model on the same axes, and comment on the fit.

First graph the data and the model on the same axes. Choose the correct graph below.

☐ A.☐ B.☐ C.☐ D.

All graphs are shown in $[0, 50]$ by $[0, 20]$ with $X_{\text{scl}}=5$ and $Y_{\text{scl}}=2$ viewing rectangles.

Comment on the fit of the model on the data. Choose the correct answer below.

- ☐ A. The model is a poor fit because it does not pass close to any of the data points.
- ☐ B. The model is a mediocre fit because it passes close to some of the data points.
- ☐ C. The model is a good fit because it passes close to all of the data points.
- ☐ D. The model is an excellent fit because it passes through all of the data points.

d. How many citizens does the model predict will have Alzheimer's disease or other dementia in 2055?

million

(Type an integer or decimal rounded to one decimal place.)

e. During what year after 2000 does the model predict that 9.9 million citizens will have Alzheimer's or other dementia?

2. The total revenue function for a product is given by $R = 690x$ dollars, and the total cost function for this same product is given by $C = 18,000 + 60x + x^2$, where C is measured in dollars. For both functions, the input x is the number of units produced and sold.
- Form the profit function for this product from the two given functions.
 - What is the profit when 22 units are produced and sold?
 - What is the profit when 36 units are produced and sold?
 - How many units must be sold to break even on this product?

a. Write the profit function.

$$P(x) = \boxed{}$$

(Simplify your answer.)

b. When 22 units are produced and sold, there is a

(1) _____ of \$ $\boxed{}$.

(Simplify your answer.)

c. When 36 units are produced and sold, there is a

(2) _____ of \$ $\boxed{}$.

(Simplify your answer.)

d. The number of units that must be sold to break even on this product is $\boxed{}$.

(Simplify your answer. Use a comma to separate answers as needed.)

- (1) ☐ profit (2) ☐ loss
☐ loss ☐ profit

3. The number of millions of people age 12 and older who used marijuana during the years 2003 to 2008 is described by the function $M(x) = -0.069(x - 4.5)^2 + 25.6$ for $3 \leq x \leq 8$, where x is the number of years after 2000.

- The graph of this function is a shifted graph of which basic function?
- Find and interpret $M(6)$.
- Sketch a graph of $y = M(x)$ for $3 \leq x \leq 8$.

a. Choose the correct answer below.

- ☐ $f(x) = |x|$
☐ $f(x) = x^3$
☐ $f(x) = x^2$
☐ $f(x) = \sqrt{x}$

b. Find $M(6)$.

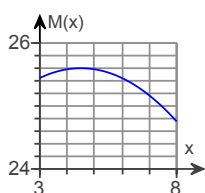
$$M(6) = \boxed{}$$

(Type an integer or a decimal rounded to three decimal places as needed.)

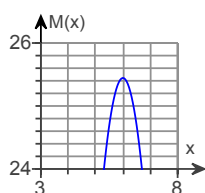
This means that in $\boxed{}$, the number of people 12 and older who used marijuana was $\boxed{}$.

c. Choose the correct graph of the function for $3 \leq x \leq 8$ below

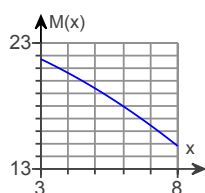
☐ A.



☐ B.



☐ C.



☐ D.

